

SPIRE

A Contested-Logistics Decision Engine for the Stand-in Force

BLUF. A 3d MLR S-4 forward in the First Island Chain today cannot see Class I, V, and VIII in a single pane — those classes live in three incompatible systems while GCSS-MC focuses on Class IX requisitioning. SPIRE collapses GCSS-MC, DRRS-MC, and forward sensor feeds into one role-shaped view that runs offline on a 2U box in a CONEX, with tamper-evident audit and one-click coalition release. **TRL 4, validated at Modern Day Marine 2026 (1st place, 30 APR 2026). Operational pilot is the next step.**

The Operator's Problem

Today, a forward unit taking a Class VIII demand spike under degraded SATCOM has three systems open and a phone call. The planner has minutes — not hours — to choose between cross-level, expedite, or emergency requisition. The decision is unauditably; the artifact is a spreadsheet.

Capability Gap

“GCSS-MC now primarily focuses on requisitioning Class IX repair parts.” Classes I, V, and VIII are managed by “three separate incompatible systems.” “Conventional sustainment remain[s] the SIF concept’s greatest vulnerability.”

— *Sustainment of the Stand-in Force*, War on the Rocks (Sept 2022). The same fragmentation is acknowledged in the *Marine Corps Concept for Logistics* (Dec 2024).

Approach

- **Universal Ingest.** 8 formats (CSV, TSV, XLSX, JSON, JSONL, XML, fixed-width, EDI X12) across 6 channels (drop, SFTP, IMAP, HTTP, DB CDC, Kafka). 4 pre-built adapters: GCSS-MC ECP / UTIL / SR-Header, DRRS-MC C-Rating. A local Gemma 4 26B model proposes the column map for non-canonical sources; mappings are saved per unit.
- **Tamper-evident audit chain.** Every state change is SHA-256 hash-chained and Ed25519-signed; exports as CEF for Splunk / ArcSight / QRadar. Inspector verification is offline, against a published key.
- **Local-first.** Runs on a 2U CONEX box. Vector-clock conflict resolution reconciles forward-node writes when SATCOM restores. No cloud egress.
- **Coalition-ready.** Live partner-scoped redaction (FVEY, JPN, AUS, PHL). One-click ZIP with manifest hash and release ID.

Operational Impact

- **One pane across Class I / V / VIII / IX** — eliminates the cross-system reconciliation step that consumes the forward planner’s first ten minutes of every Class VIII / V surge.
- **Forward decisions are auditable.** The decision an S-4 made at H+0 is verifiable against the public key at H+30 days.
- **No SATCOM dependency for forward planning** — aligns with the SIF requirement that sustainment be *“light, flexible, responsive, resilient, and redundant”* (HQMC, *A Concept for Stand-in Forces*, Dec 2021).

Maturity & Risk

TRL 4. Validated against synthetic GCSS-MC ECP / UTIL / SR-Header data at MDM 2026; hash-chained audit verified offline. **TRL 5 requires** a sanitized real GCSS-MC pull-through from a sponsoring MEF G-4 or MLR S-4. ATO is not yet established; architecture is designed-in for IL5 (audit chain, hierarchical RBAC, CAC/PIV scaffolding, no cloud egress) but unaccredited.

Why Now

The 39th Commandant has named logistics the **pacing function**. The *Marine Corps Concept for Logistics* (Dec 2024) calls for resilience through redundancy, dispersion, and innovation. The data-fragmentation gap is named in published doctrine. The window to put SPIRE in operator hands is the FY27 POM cycle.

Path Forward — Ask

1. **90-day operational pilot** with one MEF G-4 or MLR S-4 willing to point a sanitized GCSS-MC export at the ingest path.
2. **An MCWL evaluator** to red-team SENTRY against operator-malpractice patterns.
3. **A POC at MCSC PM-MC2I or DC I&L** to translate operator requirements into a contract vehicle (SBIR Phase II or OTA).

POC: Jesse Morgan · Thornveil · jesse@thornveil.ai

Repo: jeranaiaas/spire (private; access on request)

Pilot install: docker-compose up — under 5 min, clean laptop to live demo

v1.1 · 04 MAY 2026 · Distribution unrestricted by author